

Generalization-Aware AIS Trajectory Prediction and Risk-Warning Benchmark

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Submission-readiness note: this draft is synchronized with the current evidence pack. The readiness audit currently reports `submission_ready_candidate`. If blocking gaps are listed below, the manuscript is a submission draft rather than a final submission.

Abstract

Short-term vessel trajectory prediction is often evaluated with increasingly complex neural models, but maritime decision-support systems also require auditable data protocols, strong kinematic baselines, generalization tests, and downstream risk-warning evidence. This study presents a reproducible AIS trajectory-prediction pipeline using NOAA MarineCadastre.gov data [`@noaa_marinecadastre_ais`; `@noaa_digitalcoast_marinecadastre`]. The current evidence pack keeps vessel identifiers, source dates, timestamps, regions, speed, and turning-intensity metadata so that temporal and vessel-disjoint splits can be evaluated. In the current artifact run, the processed dataset contains 186326 trajectory windows from 7425 MMSI values and source dates 2024-01-02, 2024-01-09, 2024-02-06, 2024-03-05. The best temporal-test ADE in the current run is 1759.7 m from `kalman_filter_cv`, while the best vessel-disjoint ADE is 3109.4 m from `kalman_filter_cv`. A downstream AIS-derived risk-warning evaluation estimates warning precision, recall, false alarms, missed warnings, and CPA error without claiming autonomous collision avoidance.

1. Introduction

AIS trajectory prediction is useful only when model comparisons survive simple baselines and when the forecasting errors can be connected to operational quantities such as closest point of approach and warning lead time. The objective of this manuscript is therefore not to assert that a neural architecture wins by default. Instead, it asks whether a transparent AIS evidence pipeline can support stronger claims about generalization and risk-warning behavior than a single-day benchmark.

2. Data Protocol

The dataset builder records raw file checksums, row counts, MMSI counts, timestamp ranges, geographic bounds, processed-file checksum, split policy, and scenario-slice metadata. Each trajectory window uses 30 one-minute history steps to forecast 15 one-minute future positions. Coordinates are WGS84 latitude/longitude, and displacement metrics are Haversine meters.

The current processed artifact contains 186326 samples, 7425 unique MMSI values, and source dates 2024-01-02, 2024-01-09, 2024-02-06, 2024-03-05. Temporal split counts are `{'train': 130428, 'val': 27948, 'test': 27950}`; vessel-disjoint split counts are `{'train': 130018, 'val': 28597, 'test': 27711}`. Regions represented in the current artifact are `east_gulf_coast`, `hawaii_pacific`, `other`, `west_coast`.

3. Models

The benchmark includes constant velocity, constant acceleration, a Kalman-style constant-velocity smoother, ridge least squares, ordinary least squares, LSTM [hochreiter1997lstm], GRU [cho2014gru], and Transformer baselines [vaswani2017attention], plus a temporal convolutional network baseline [bai2018tcn]. Neural models are implemented in PyTorch [paszke2019pytorch]. Linear models are included because simple statistical baselines can be more robust than untuned deep sequence models on short-horizon AIS windows.

4. Evaluation Design

The primary trajectory metrics are ADE, FDE, RMSE, and MAE in meters. Generalization is evaluated with both temporal holdout and vessel-disjoint holdout, using aligned per-sample errors for statistical tests. Scenario slices are produced by region, speed bin, and turning-intensity bin.

Risk-warning evaluation uses observed future pairwise separation to define AIS-derived warnings within the forecast horizon. Predicted trajectories are then evaluated for warning precision, recall, false alarms, missed warnings, lead-time error, and CPA error. This is decision-support evidence, not a validated closed-loop collision-avoidance system under COLREGs [imo_colregs].

5. Trajectory Results

Split	Model	Status	ADE (m)	FDE (m)	RMSE (m)	MAE (m)	Train	Val	Test
temporal_test	constant_acceleration	ok	35076.6	76360.6	737164.2	28836.3	130428	27948	27950
temporal_test	constant_velocity	ok	2751.3	4469.8	77190.7	1742.6	130428	27948	27950
temporal_test	gru_baseline	ok	25215.5	25571.8	29923.9	14863.6	130428	27948	27950
temporal_test	kalman_filter_cv	ok	1759.7	2704.5	22422.7	1101.1	130428	27948	27950
temporal_test	linear_lstsq	ok	3052.3	4908.7	13349.1	1910.3	130428	27948	27950
temporal_test	lstm_baseline	ok	36039.2	36116.8	57826.3	22176.2	130428	27948	27950
temporal_test	ridge_lstsq	ok	3141.7	5079.5	12468.9	1965.8	130428	27948	27950
temporal_test	tcn_baseline	ok	47095.5	47078.0	48378.2	29703.9	130428	27948	27950
temporal_test	transformer_baseline	ok	56310.7	55923.5	57766.4	34854.0	130428	27948	27950
vessel_disjoint_test	constant_acceleration	ok	36237.1	67988.2	1786906.6	55853.5	130018	28597	27711
vessel_disjoint_test	constant_velocity	ok	9553.5	17014.2	198051.0	6181.8	130018	28597	27711
vessel_disjoint_test	gru_baseline	ok	23989.0	24569.4	27376.0	14911.1	130018	28597	27711
vessel_disjoint_test	kalman_filter_cv	ok	3109.4	5979.6	47635.6	1941.5	130018	28597	27711
vessel_disjoint_test	linear_lstsq	ok	8113.3	14869.8	172502.5	5291.8	130018	28597	27711
vessel_disjoint_test	lstm_baseline	ok	51010.6	50890.1	77725.3	32747.2	130018	28597	27711
vessel_disjoint_test	ridge_lstsq	ok	3446.9	6463.4	18536.5	2153.1	130018	28597	27711
vessel_disjoint_test	tcn_baseline	ok	32833.1	34379.1	34775.3	20214.3	130018	28597	27711
vessel_disjoint_test	transformer_baseline	ok	47559.1	45907.0	46592.6	30158.9	130018	28597	27711

6. Risk-Warning Results

The current risk-warning artifact evaluates 2000 AIS-derived encounter scenarios from 5000 evaluation samples, using a 0.5 nmi warning threshold and 3.0 nmi search radius.

Model	Scenarios	Precision	Recall	False alarm rate	Missed warning rate	Mean abs CPA error (nmi)
constant_velocity	2000	0.961	0.894	0.013	0.106	0.082
kalman_filter_cv	2000	0.963	0.900	0.012	0.100	0.092
linear_lstsq	2000	0.724	0.871	0.117	0.129	0.346

7. Reproducibility and Artifact Synchronization

The synchronized artifact sources are:

- outputs/audit/multiday_data_manifest.json
- outputs/audit/multiday_split_manifest.csv
- outputs/final_multiday/model_metrics.csv
- outputs/final_multiday/neural_tuning_protocol.json
- outputs/final_multiday/generalization_metrics.csv
- outputs/final_multiday/statistical_tests.json
- outputs/final_risk/risk_metrics.json
- outputs/final_risk/risk_scenarios.csv
- outputs/latest_predictions/prediction_manifest.json
- outputs/final_submission/readiness_report.json

The high-quality roadmap pipeline command is:

```
PYTHON_BIN=.venv/bin/python bash scripts/run_high_quality_pipeline.sh
```

8. Limitations

- No blocking readiness gaps are currently reported by the automated audit.

- Risk-warning results should not be interpreted as autonomous collision-avoidance validation.
- Model rankings reflect the current stratified time-block protocol; full-day or alternative time-block protocols should be run before making broader all-day traffic claims.
- Neural tuning is documented as a validation-set proxy search; strong architecture-superiority claims require broader search evidence and independent external validation.

9. Conclusion

The project now provides a synchronized path from AIS data audit to trajectory metrics, generalization analysis, operational risk-warning evaluation, latest-data offline prediction, and manuscript artifacts. The current repository state should be treated according to the readiness audit: if blockers remain, the artifact is a high-quality-journal draft package rather than a submission-ready paper.

References

References are stored in `paper/references.bib`.